Data Visualization Onboarding Project Specification Sheet

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1. Introduction

1.1 Summary

Up until the 2026/2027 season, VM has not had a centralized means of accessing vehicle data for analysis and iteration. The VM team needs an easily accessible, maintainable, and scalable interface to read data from the vehicle. The solution should be accessible via computer or tablet and handle pulling past data along with real-time data. The solution should have a presentable dashboard to easily interpret sensor information from the vehicle. The creation of a data visualization software would expedite the process of iteration for future vehicles on VM and assist other sub-teams in their work.

1.2 Context

Going into the new season, VM is preparing to run its first electric vehicle at competition. It is imperative that the necessary logging tools are completed before the vehicle runs. In order to begin the process of iteration, there needs to be easy-access software for members to analyze the performance of the car from past runs or in real-time.

1.3 Goals and Technical Requirements

- The solution must be able to read from a CSV file and present both numerical and visual representations of all data points. Visual and numerical representations include but are not limited to:
    - Maximum, minimums, and averages for all data points
    - Graphs of all data points over time
    - Map of the car location over time
    - Movable timeline to go through each session
- The solution must be able to present all of the following data points
    - Primary outputs (direct filtered sensor data):
    - G-G scatterplot (lateral and longitudinal acceleration)
    - Steering trace vs. time
    - Accelerator/brake pressure trace vs. time
    - Speed vs. track position
    - Yaw velocity vs. time
    - Yaw acceleration vs. time

    - F+R slip angle vs. time
    - F+R slip ratio vs. time
    - Predicted Fy vs. actual Fy vs. time (hit me up for help with this one, requires tyre model)
    - Vehicle sideslip angle vs. time
    - Vehicle roll gradient vs. time
    - Vehicle pitch gradient vs. time
- The solution must be able to compare different sessions of the car to indicate differences between data points.
- The solution must be able to be accessed via computer or tablet
- The solution must have an interface to label and store past runs
- The solution must be able to track data points in real-time
- The solution should have a login page
- The solution should be customizable to only present certain information (insertable and removable blocks)